North Sea banks where big seas break

Seabed: EMODnet Bathymetry (about 500 m grid, depths below Lowest Astronomical Tide). Wave history: ERA5 at 22 points in the southern North Sea, 2000-01-01 to 2024-11-18. Forecast: Open-Meteo 2026-10-17 00:00 (Tp = 1.13 x mean period).

What this shows

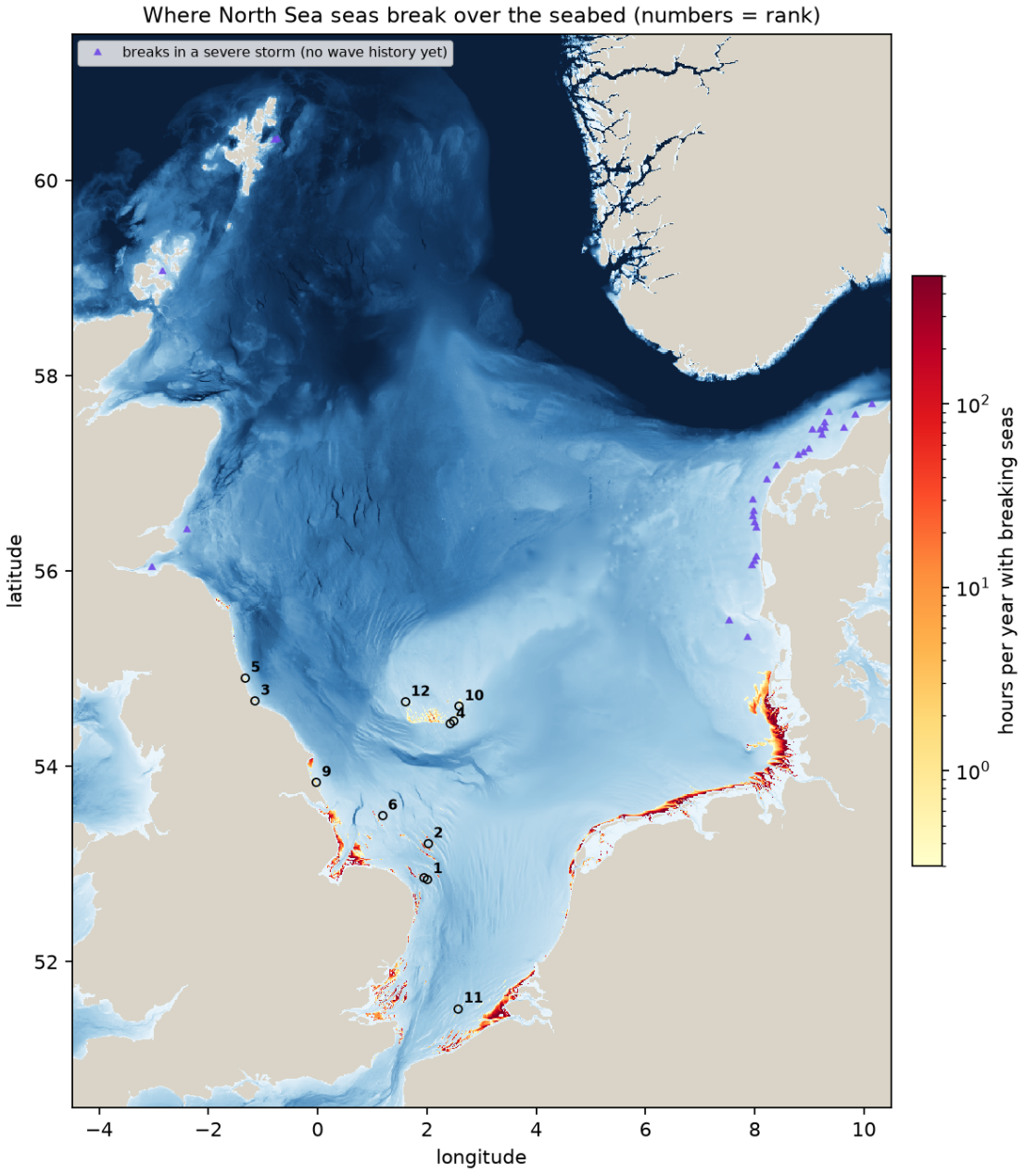
• In the southern North Sea, where 26 years of wave history were downloaded, 182 banks and shoals break in big seas at least once. 61 of them are deeper than 8 m at low tide, so ships cross them routinely, and those are where breaking seas catch vessels out.

• Across the rest of the North Sea a further 29 banks would break in a severe storm (Hs 9 m at 12 s). They are ranked by how small a sea sets them off, since their wave history isn't downloaded yet.

• The sea starts to break over a bank when the significant wave height reaches roughly half to six tenths of the water depth. Longer-period seas feel the bottom sooner and break at a lower height, so a 12 to 14 s northerly swell after a storm is more dangerous over a bank than a short, steep 8 s wind sea of the same height.

• The most dangerous navigable banks are Hammond Knoll, Shoal 57 km off Cromer, Shoal 4 km off Hartlepool.

• In the current forecast (Open-Meteo 2026-10-17 00:00 (Tp = 1.13 x mean period)), 53 banks reach breaking level, worst Shoal 19 km off Terschelling around 2026-10-17 21:00 UTC.



Southern North Sea, ranked: navigable banks (deeper than 8 m at low tide)

#	Bank	Min depth (m LAT)	Breaking h/yr	Storms /yr	Hs to break Tp 10 s	Hs to break Tp 13 s	Seas from	Forecast
1	Hammond Knoll	8	30	2.6	4.2 m	3.9 m	NNW, N	none
2	Shoal 57 km off Cromer	9	24	2.3	4.3 m	4.1 m	NNW, N	none
3	Shoal 4 km off Hartlepool	9	13	0.9	4.8 m	4.4 m	N, E, ENE	none
4	Dogger Bank, central	11	17	2.2	5.3 m	5.2 m	NNW, NW	none
5	Shoal 14 km off Tynemouth	9	12	0.9	4.8 m	4.4 m	N, E, ENE	none
6	Shoal 63 km off Cromer	9	14	0.0	4.7 m	4.7 m	NNW, N	none
7	Hammond Knoll (2)	9	13	1.4	4.7 m	4.4 m	NNW	none
8	Dogger Bank, central (2)	11	8	1.3	5.6 m	5.6 m	NNW, NW	none
9	Shoal 30 km off Spurn Head	9	7	0.7	4.3 m	4.0 m	N, ENE, E	none
10	Dogger Bank, central (3)	12	8	1.3	5.8 m	6.0 m	NNW, NW, SSW	none
11	Shoal 41 km off Ostend	10	5	1.0	5.1 m	4.8 m	SW, NNW	none
12	Dogger Bank, west	12	6	1.0	5.7 m	5.7 m	NNW, NW, ENE	none
13	Thornton Bank	10	4	0.8	5.2 m	5.1 m	SW, NNW, NW	none
14	Dogger Bank, central (4)	12	6	1.0	6.0 m	6.1 m	NNW, SSW, NW	none
15	Shoal 12 km off IJmuiden	9	8	1.1	4.3 m	4.0 m	NW, WNW, W, WSW	none
16	Hammond Knoll (3)	10	3	0.6	5.2 m	4.9 m	NNW	none
17	Swarte Bank	11	3	0.6	5.3 m	5.1 m	NNW, N	none
18	Dogger Bank, west (2)	12	3	0.6	5.9 m	6.0 m	NNW, NW	none
19	Dogger Bank, central (5)	12	3	0.7	5.9 m	6.0 m	NNW, NW	none
20	Shoal 42 km off Vlissingen	11	1	0.2	5.5 m	5.4 m	SW, NW, WSW, W	none
21	Galloper Bank	9	2	0.3	4.5 m	4.2 m	SSW, NNW	none
22	Hammond Knoll (4)	11	3	0.5	5.3 m	5.1 m	NNW, N	none
23	Shoal 47 km off Skegness	8	2	0.2	4.0 m	3.9 m	N, NNE	none
24	Shoal 33 km off Skegness	8	2	0.2	4.1 m	4.0 m	N, NNE, NNW, NE	none
25	Shoal 36 km off Sylt	12	1	0.2	6.1 m	6.4 m	W, WNW, NW	none
26	Shoal 44 km off Cromer	11	2	0.4	5.4 m	5.3 m	NNW, SSW	none
27	Dogger Bank, central (6)	14	1	0.2	6.7 m	7.0 m	SSW, NNW, WSW, WNW	none
28	Shoal 69 km off Spurn Head	12	2	0.3	5.8 m	5.9 m	NNW, N, ENE, E	none
29	Dogger Bank, west (3)	13	1	0.3	6.3 m	6.5 m	NNW, WNW, SSW	none
30	Dogger Bank, central (7)	13	2	0.4	6.5 m	6.8 m	SSW, NNW, W, NW	none

Hs is the significant wave height of the open sea near the bank (what a forecast gives you); Tp is the peak period. '-' means no realistic sea at that period is big enough. Breaking h/yr is the average number of hours per year that at least 1 in 20 waves breaks over the shallowest part. Depths are at lowest astronomical tide: at high water or in a positive storm surge the triggers rise by about half the extra depth.

Southern North Sea: shallow sandbanks (3 to 8 m at low tide)

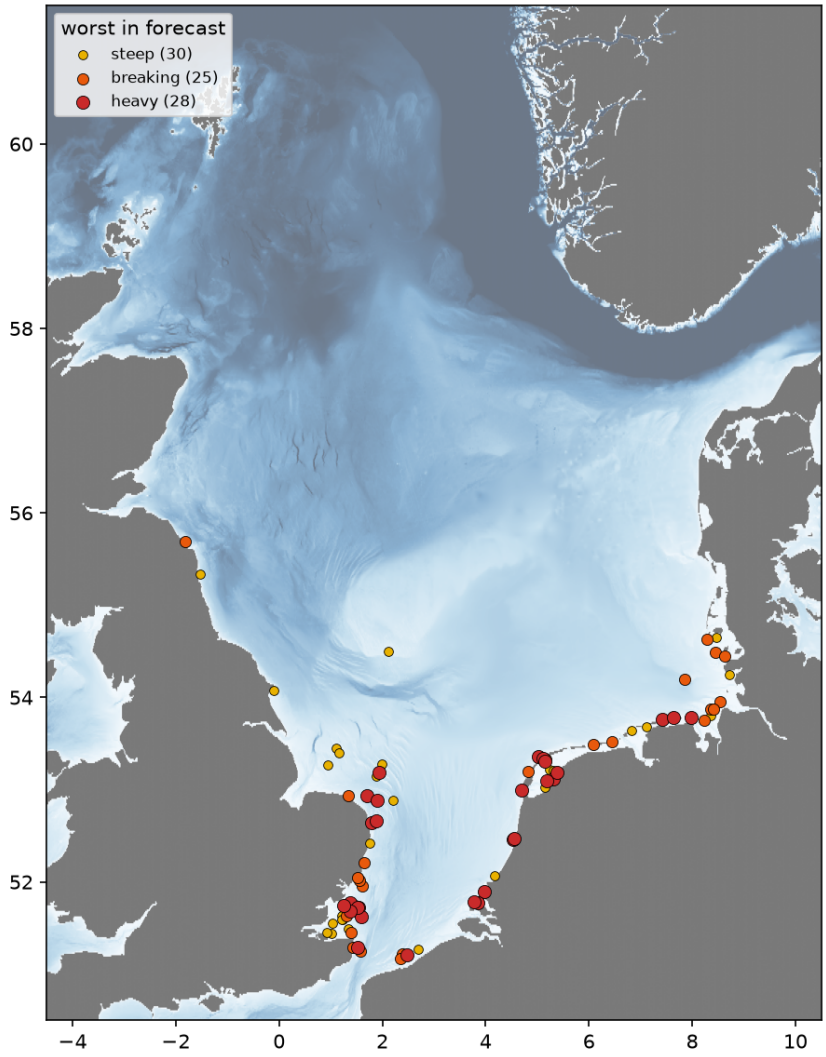
#	Bank	Min depth (m LAT)	Breaking h/yr	Storms /yr	Hs to break Tp 10 s	Hs to break Tp 13 s	Seas from	Forecast
62	Shoal 19 km off Terschelling	3	2598	40.1	1.3 m	1.2 m	SW, WSW, W, NW	heavy
63	Shoal 30 km off Sylt	3	2248	41.3	1.4 m	1.3 m	NW, W, WNW, WSW	breaking
64	Shoal 31 km off Scheveningen	3	2007	29.7	1.3 m	1.2 m	WSW, W, SW, NW	heavy
65	Kentish Knock	3	1508	49.1	1.3 m	1.2 m	SW, SSW	heavy
66	Shoal 42 km off Borkum	3	2058	44.5	1.3 m	1.2 m	NNW, NW, W, WNW	breaking
67	Kentish Knock (2)	3	1469	47.0	1.3 m	1.2 m	SSW, SW	heavy
68	Shoal 3 km off Helgoland	4	1050	29.9	1.8 m	1.6 m	NW, W, WNW, WSW	breaking
69	Shoal 51 km off Cromer	3	1577	43.5	1.5 m	1.4 m	NNW, N	heavy
70	Shoal 22 km off Borkum	3	1811	33.9	1.5 m	1.4 m	NNW, NW, WNW	none
71	Sunk Sand	3	1247	44.9	1.3 m	1.3 m	SW, SSW	heavy
72	Kentish Knock (3)	3	1384	47.1	1.4 m	1.3 m	SSW, SW	heavy
73	Scroby Sands	3	1204	40.7	1.3 m	1.2 m	N, S	heavy
74	Haisborough Sand	3	1345	40.8	1.3 m	1.2 m	NNW, N	heavy
75	Shoal 19 km off Borkum	3	1884	40.4	1.4 m	1.3 m	NNW, NW, W, WNW	breaking
76	Shoal 26 km off Den Helder	3	1384	37.2	1.4 m	1.2 m	SW, WSW, W, NW	breaking
77	Shoal 10 km off Borkum	4	1838	41.6	1.6 m	1.6 m	NW, W, WSW, WNW	steep
78	Shoal 47 km off Sylt	3	1794	38.7	1.6 m	1.6 m	NW, W, WNW, WSW	breaking
79	Shoal 15 km off Berwick	3	1388	31.7	1.5 m	1.3 m	N, NNE	breaking
80	Long Sand	3	1061	39.0	1.3 m	1.3 m	SSW, S, NE	heavy
81	Shoal 51 km off Borkum	3	1530	39.5	1.5 m	1.4 m	NW, W, WNW, NNW	heavy
82	Shoal 15 km off Skegness	3	951	4.5	1.3 m	1.2 m	ENE, E	none
83	Shoal 11 km off Terschelling	3	1125	32.8	1.4 m	1.4 m	W, NW, NNW, WNW	breaking
84	Long Sand (2)	3	1040	38.5	1.4 m	1.3 m	SW, SSW, S	breaking
85	Sunk Sand (2)	3	1014	42.2	1.5 m	1.5 m	SSW, SW	breaking
86	Shoal 28 km off Borkum	3	1037	32.5	1.3 m	1.2 m	NW, NNW	steep

Rest of the North Sea: banks ranked by the sea state that breaks them (physics only)

#	Bank	Min depth (m LAT)	Breaking h/yr	Storms /yr	Hs to break Tp 10 s	Hs to break Tp 13 s	Seas from	Forecast
183	Shoal 38 km off Lerwick	4	-	-	1.8 m	1.6 m	-	none
184	Shoal 35 km off Lerwick	5	-	-	2.3 m	2.1 m	-	none
185	Shoal 13 km off Kirkwall	5	-	-	2.5 m	2.3 m	-	none
186	Shoal 27 km off St Andrews	6	-	-	3.3 m	2.9 m	-	none
187	Shoal 24 km off Hirtshals	7	-	-	3.6 m	3.5 m	-	none
188	Shoal 58 km off Esbjerg	9	-	-	4.7 m	4.6 m	-	none
189	Shoal 40 km off Esbjerg	13	-	-	6.3 m	6.4 m	-	none
190	Shoal 33 km off Dunbar	13	-	-	6.6 m	6.6 m	-	none
191	Shoal 15 km off Hanstholm	14	-	-	7.1 m	7.2 m	-	none
192	Shoal 41 km off Hirtshals	15	-	-	7.2 m	7.4 m	-	none
193	Shoal 36 km off Hirtshals	15	-	-	7.5 m	7.5 m	-	none
194	Shoal 27 km off Thyboron	16	-	-	7.9 m	8.2 m	-	none
195	Shoal 14 km off Hvide Sande	16	-	-	7.7 m	8.2 m	-	none
196	Shoal 17 km off Hirtshals	16	-	-	8.2 m	8.1 m	-	none
197	Shoal 14 km off Hvide Sande (2)	17	-	-	8.0 m	8.4 m	-	none
198	Shoal 13 km off Hanstholm	17	-	-	8.1 m	8.5 m	-	none
199	Shoal 16 km off Thyboron	17	-	-	8.1 m	8.4 m	-	none
200	Shoal 25 km off Thyboron	17	-	-	8.0 m	8.5 m	-	none
201	Shoal 7 km off Hirtshals	17	-	-	8.3 m	8.5 m	-	none
202	Shoal 48 km off Hirtshals	17	-	-	8.1 m	8.7 m	-	none
203	Shoal 49 km off Hirtshals	17	-	-	8.1 m	8.7 m	-	none
204	Shoal 28 km off Hanstholm	17	-	-	8.1 m	8.8 m	-	none
205	Shoal 30 km off Thyboron	17	-	-	8.2 m	8.7 m	-	none
206	Shoal 43 km off Hirtshals	17	-	-	8.2 m	8.9 m	-	none
207	Shoal 46 km off Hanstholm	17	-	-	8.3 m	8.8 m	-	none
208	Shoal 21 km off Hanstholm	18	-	-	8.3 m	8.9 m	-	none
209	Shoal 17 km off Thyboron	18	-	-	8.4 m	8.9 m	-	none
210	Shoal 21 km off Thyboron	18	-	-	8.3 m	8.9 m	-	none
211	Shoal 19 km off Hvide Sande	18	-	-	8.4 m	9.1 m	-	none

Next six days

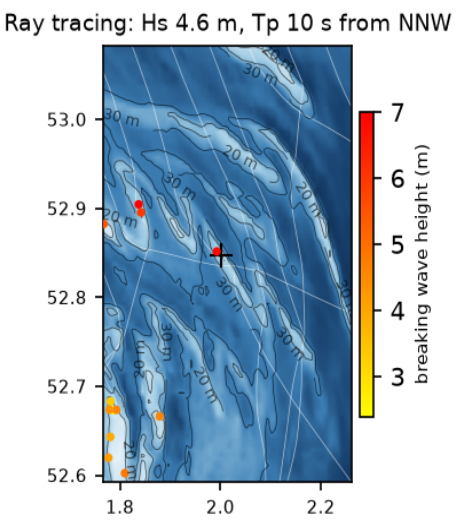
Forecast (Open-Meteo 2026-10-17 00:00 (Tp = 1.13 x mean period)): banks that will steepen or break

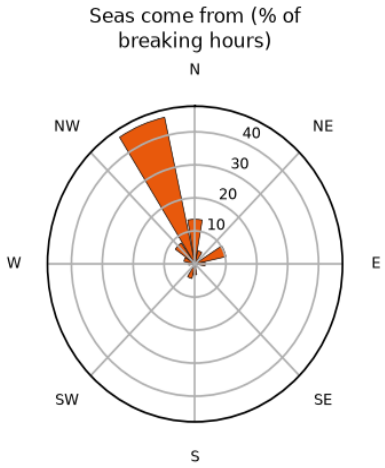


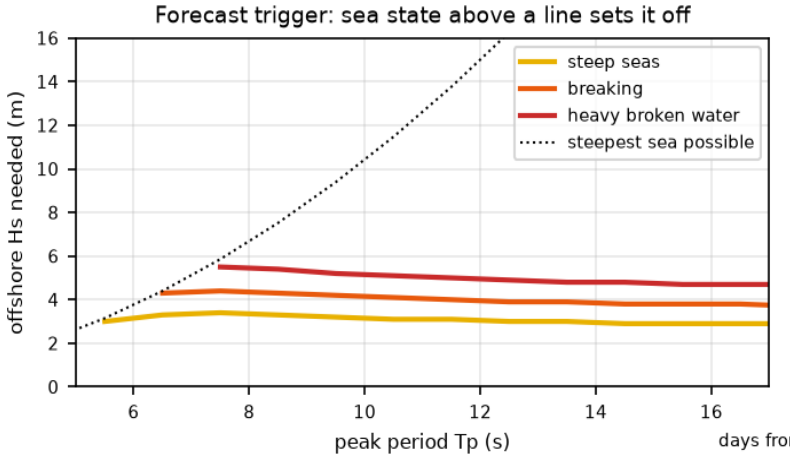
Bank	Worst level	When (UTC)	Hs	Tp	From
Shoal 19 km off Terschelling	heavy	2026-10-17 21:00	2.6 m	8 s	N
Scroby Sands	heavy	2026-10-23 12:00	3.0 m	7 s	S
Shoal 51 km off Cromer	heavy	2026-10-23 12:00	3.1 m	7 s	S
Shoal 14 km off Terschelling (2)	heavy	2026-10-17 21:00	2.6 m	8 s	N
Shoal 14 km off Terschelling	heavy	2026-10-17 21:00	2.6 m	8 s	N
Shoal 31 km off Scheveningen	heavy	2026-10-23 21:00	2.3 m	7 s	WSW
Shoal 4 km off IJmuiden	heavy	2026-10-23 21:00	2.7 m	7 s	WSW
Shoal 41 km off Vlissingen (2)	heavy	2026-10-23 21:00	2.3 m	7 s	WSW
Shoal 6 km off Den Helder	heavy	2026-10-17 21:00	2.3 m	8 s	N
Shoal 4 km off IJmuiden (2)	heavy	2026-10-23 21:00	2.7 m	7 s	WSW
Shoal 51 km off Borkum	heavy	2026-10-18 00:00	2.5 m	8 s	N
Haisborough Sand	heavy	2026-10-18 00:00	2.1 m	8 s	NNE
Shoal 46 km off Helgoland	heavy	2026-10-18 00:00	2.5 m	8 s	N
Long Sand	heavy	2026-10-18 03:00	2.0 m	7 s	NNE
Kentish Knock	heavy	2026-10-18 03:00	2.0 m	7 s	NNE

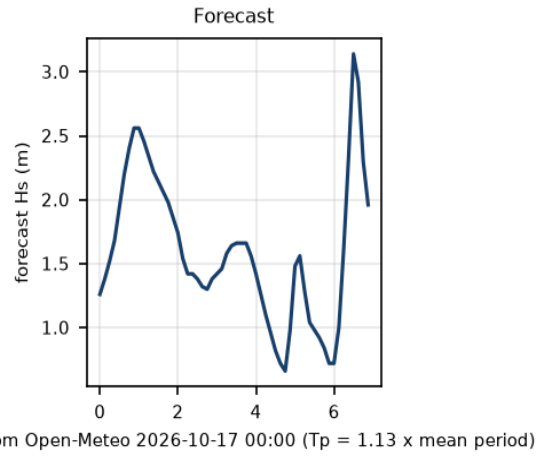
The top banks, one by one

1. Hammond Knoll (52.847 N, 2.003 E). Shallowest 8.4 m at low tide, 29 km offshore, 3 km2 of breaking ground. About 30 hours a year of breaking seas in 2.6 storms, mostly in Dec, Nov, Feb. Waves break when the offshore sea reaches Hs 4.4 m at Tp 8 s; Hs 4.2 m at Tp 10 s; Hs 4.0 m at Tp 12 s; Hs 3.9 m at Tp 14 s, from NNW, N. Worst in the record: 2020-09-25, Hs 5.9 m, Tp 10 s from NNW.

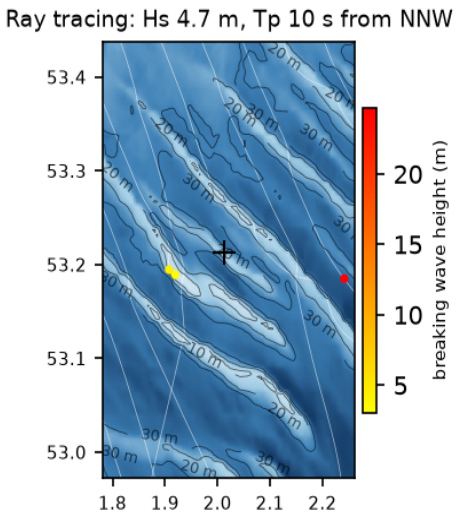


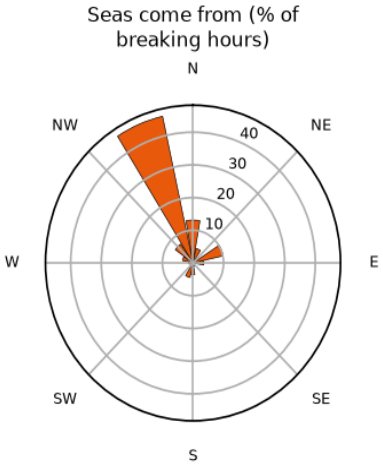


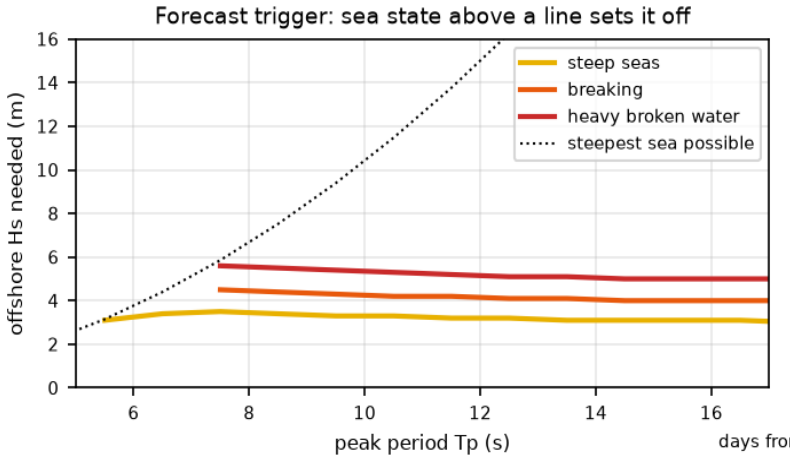


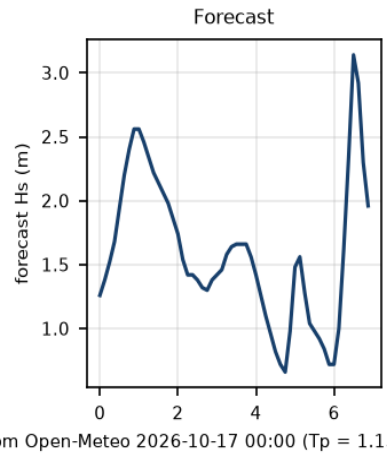


2. Shoal 57 km off Cromer (53.212 N, 2.013 E). Shallowest 8.7 m at low tide, 50 km offshore, 1 km2 of breaking ground. About 24 hours a year of breaking seas in 2.3 storms, mostly in Nov, Feb, Dec. Waves break when the offshore sea reaches Hs 4.5 m at Tp 8 s; Hs 4.3 m at Tp 10 s; Hs 4.2 m at Tp 12 s; Hs 4.1 m at Tp 14 s, from NNW, N. Worst in the record: 2020-09-25, Hs 5.9 m, Tp 10 s from NNW.

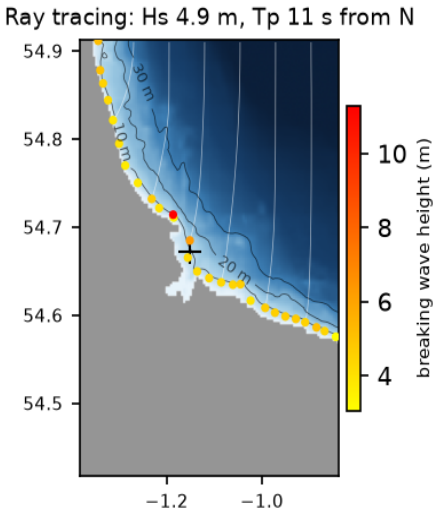


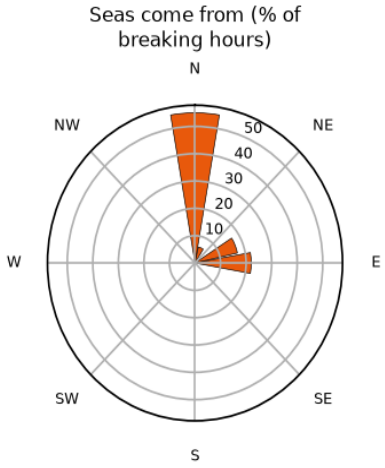


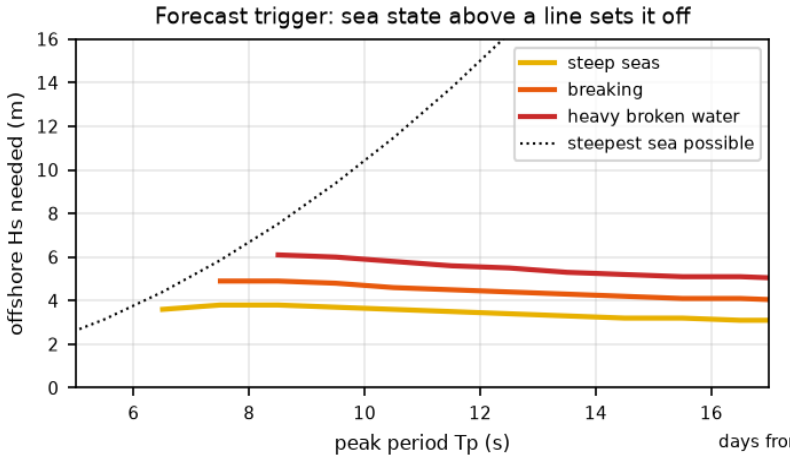


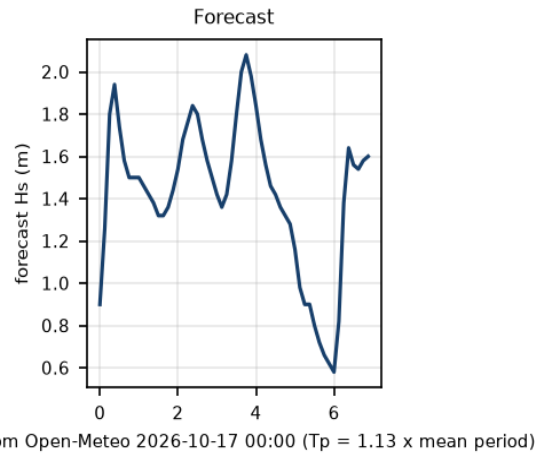


3. Shoal 4 km off Hartlepool (54.672 N, -1.152 E). Shallowest 9.3 m at low tide, 2 km offshore, 7 km2 of breaking ground. About 13 hours a year of breaking seas in 0.9 storms, mostly in Mar, Nov, Oct. Waves break when the offshore sea reaches Hs 4.9 m at Tp 8 s; Hs 4.8 m at Tp 10 s; Hs 4.5 m at Tp 12 s; Hs 4.3 m at Tp 14 s, from N, E, ENE. Worst in the record: 2021-11-27, Hs 7.2 m, Tp 12 s from N.

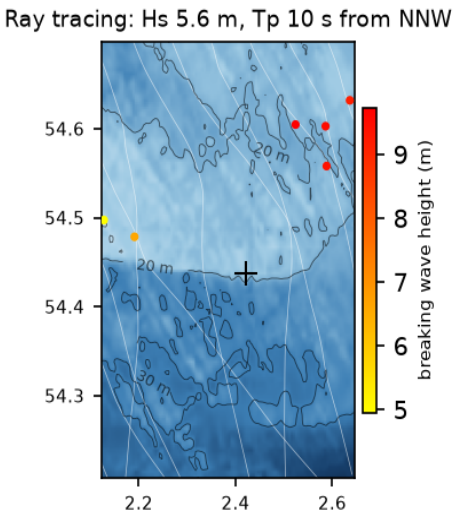


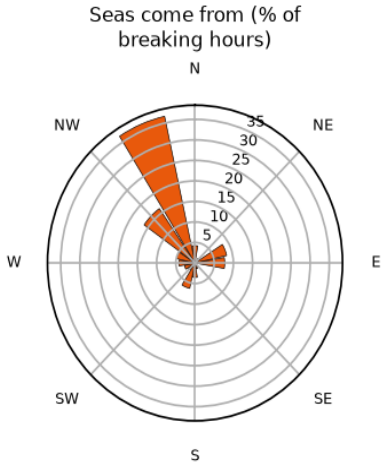


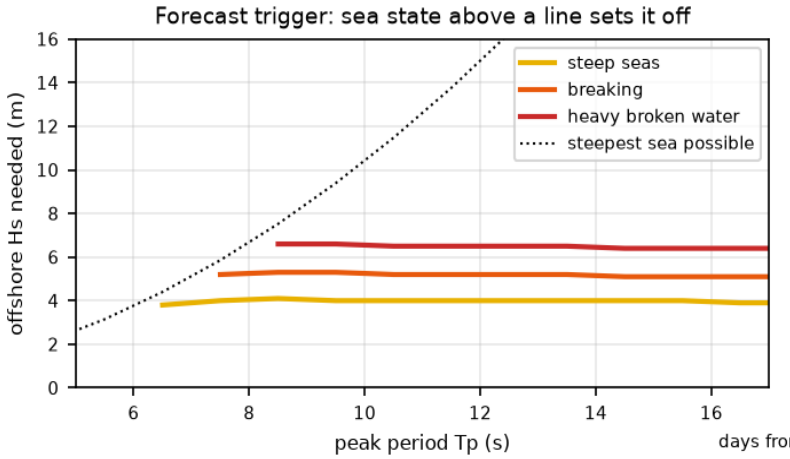


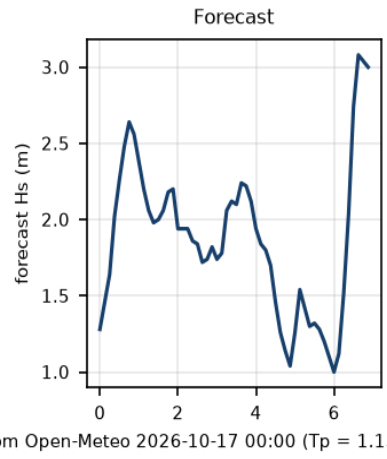


4. Dogger Bank, central (54.438 N, 2.423 E). Shallowest 10.6 m at low tide, 162 km offshore, 10 km2 of breaking ground. About 17 hours a year of breaking seas in 2.2 storms, mostly in Dec, Feb, Jan. Waves break when the offshore sea reaches Hs 5.2 m at Tp 8 s; Hs 5.3 m at Tp 10 s; Hs 5.2 m at Tp 12 s; Hs 5.2 m at Tp 14 s, from NNW, NW. Worst in the record: 2000-10-30, Hs 7.1 m, Tp 10 s from SSW.

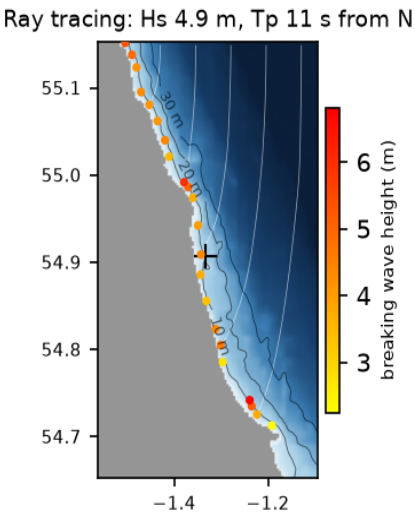


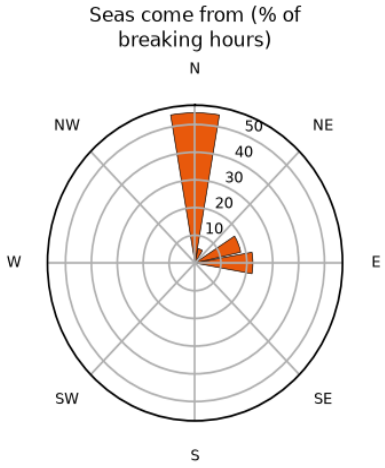


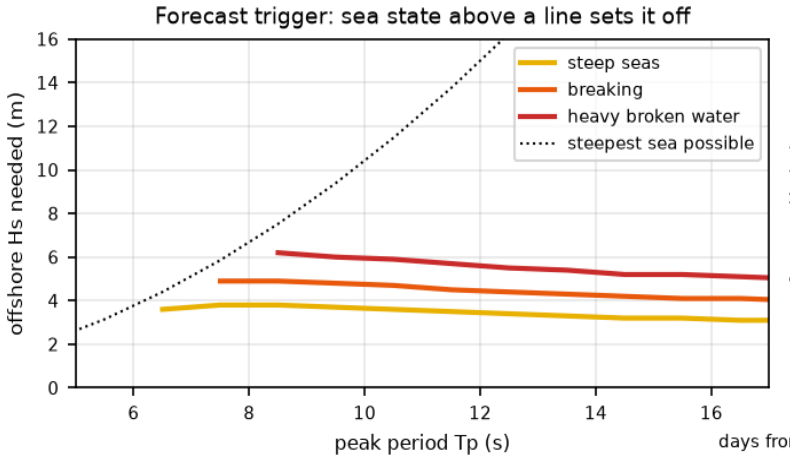


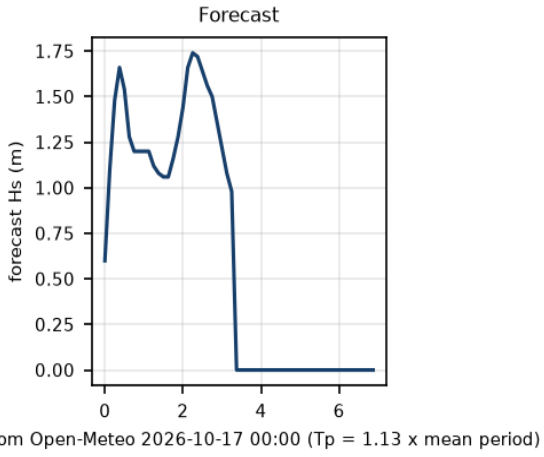


5. Shoal 14 km off Tynemouth (54.907 N, -1.333 E). Shallowest 9.3 m at low tide, 2 km offshore, 5 km2 of breaking ground. About 12 hours a year of breaking seas in 0.9 storms, mostly in Mar, Nov, Oct. Waves break when the offshore sea reaches Hs 4.9 m at Tp 8 s; Hs 4.8 m at Tp 10 s; Hs 4.5 m at Tp 12 s; Hs 4.3 m at Tp 14 s, from N, E, ENE. Worst in the record: 2021-11-27, Hs 7.2 m, Tp 12 s from N.

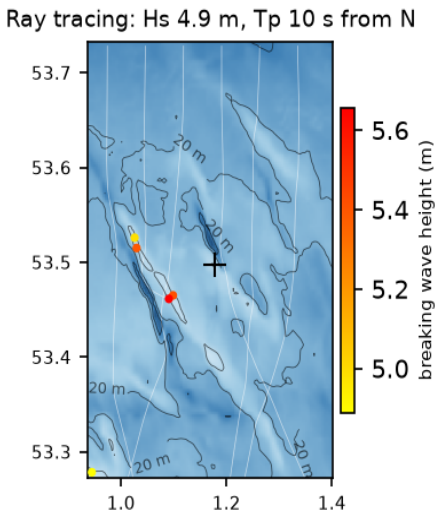


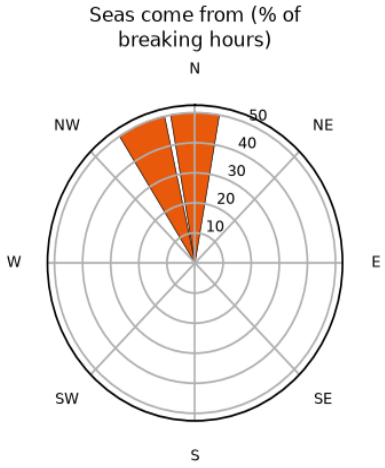


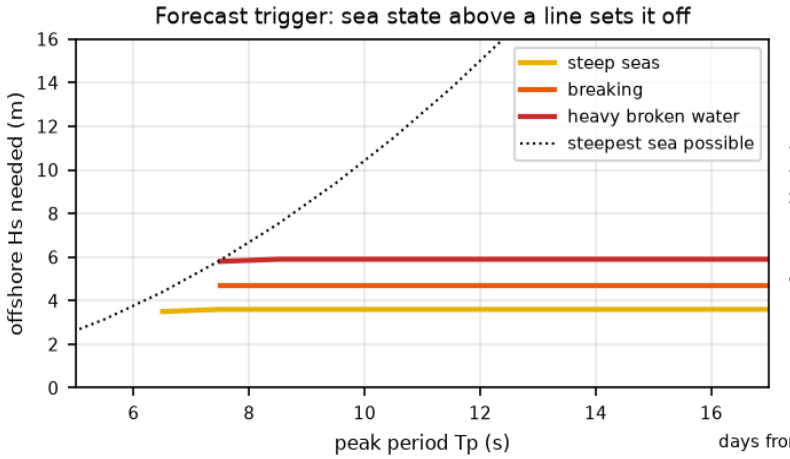


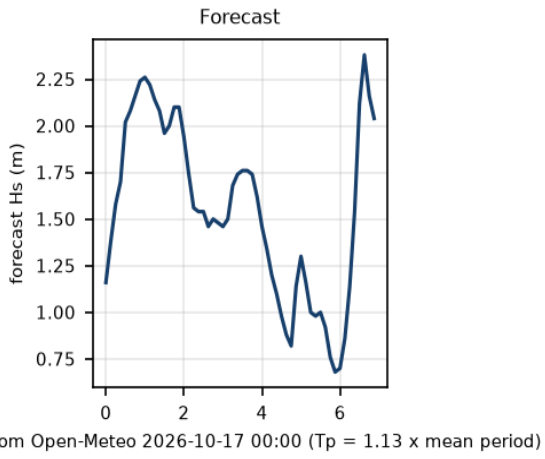


6. Shoal 63 km off Cromer (53.498 N, 1.178 E). Shallowest 9.5 m at low tide, 49 km offshore, 1 km2 of breaking ground. About 14 hours a year of breaking seas in 0.0 storms, mostly in Sep. Waves break when the offshore sea reaches Hs 4.7 m at Tp 8 s; Hs 4.7 m at Tp 10 s; Hs 4.7 m at Tp 12 s; Hs 4.7 m at Tp 14 s, from NNW, N. Worst in the record: 2020-09-25, Hs 4.9 m, Tp 10 s from NNW.

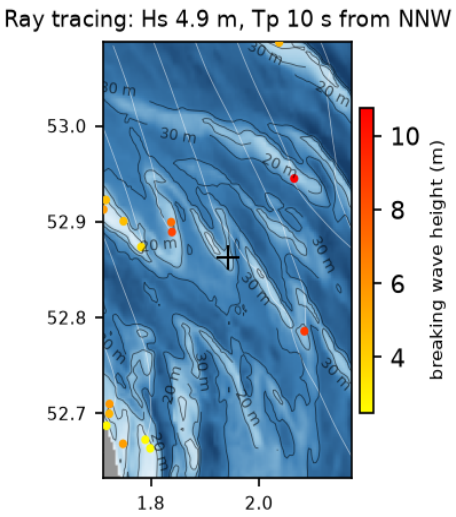


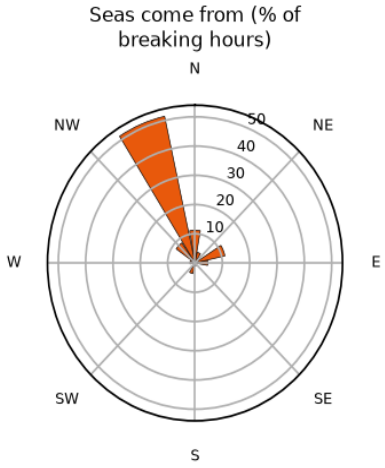


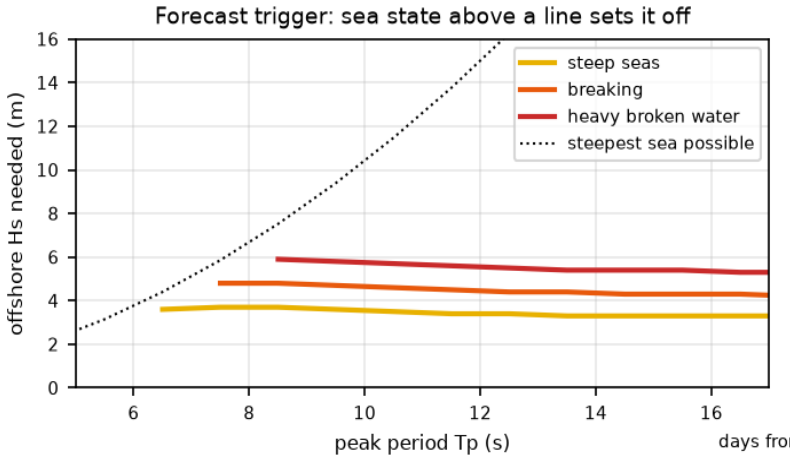


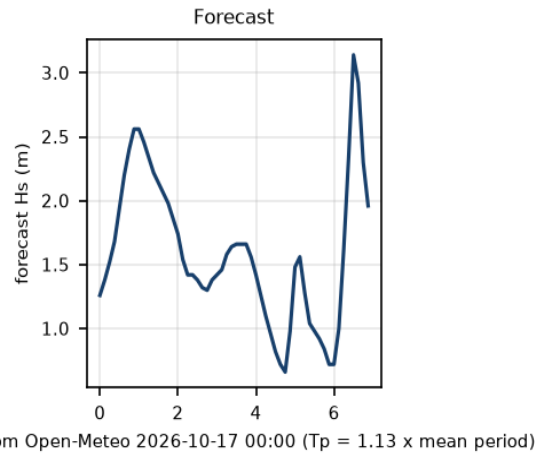


7. Hammond Knoll (2) (52.862 N, 1.942 E). Shallowest 9.3 m at low tide, 25 km offshore, 1 km2 of breaking ground. About 13 hours a year of breaking seas in 1.4 storms, mostly in Nov, Feb, Jan. Waves break when the offshore sea reaches Hs 4.8 m at Tp 8 s; Hs 4.7 m at Tp 10 s; Hs 4.5 m at Tp 12 s; Hs 4.4 m at Tp 14 s, from NNW. Worst in the record: 2020-09-25, Hs 5.9 m, Tp 10 s from NNW.

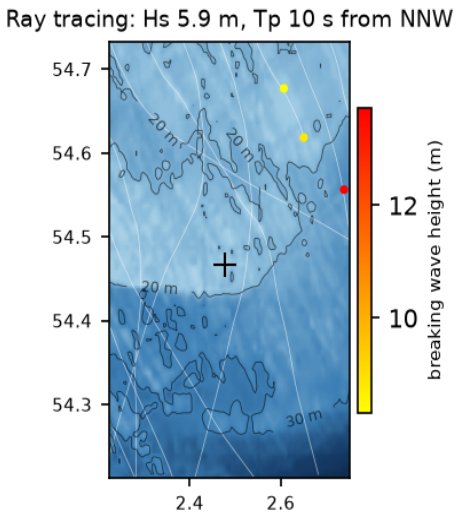


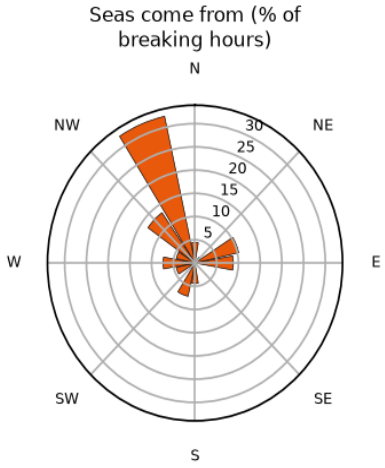


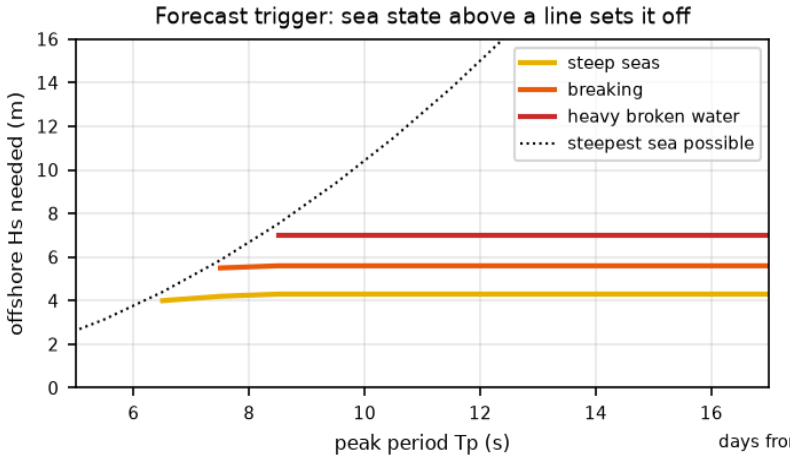


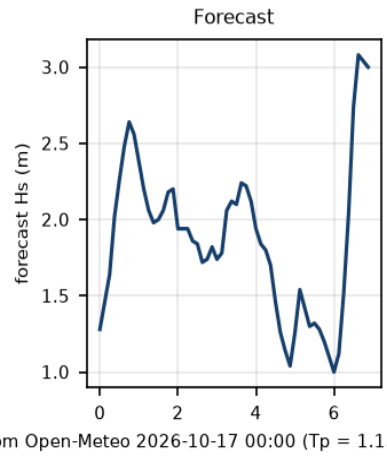


8. Dogger Bank, central (2) (54.468 N, 2.478 E). Shallowest 11.3 m at low tide, 167 km offshore, 12 km2 of breaking ground. About 8 hours a year of breaking seas in 1.3 storms, mostly in Dec, Nov, Jan. Waves break when the offshore sea reaches Hs 5.5 m at Tp 8 s; Hs 5.6 m at Tp 10 s; Hs 5.6 m at Tp 12 s; Hs 5.6 m at Tp 14 s, from NNW, NW. Worst in the record: 2000-10-30, Hs 7.1 m, Tp 10 s from SSW.



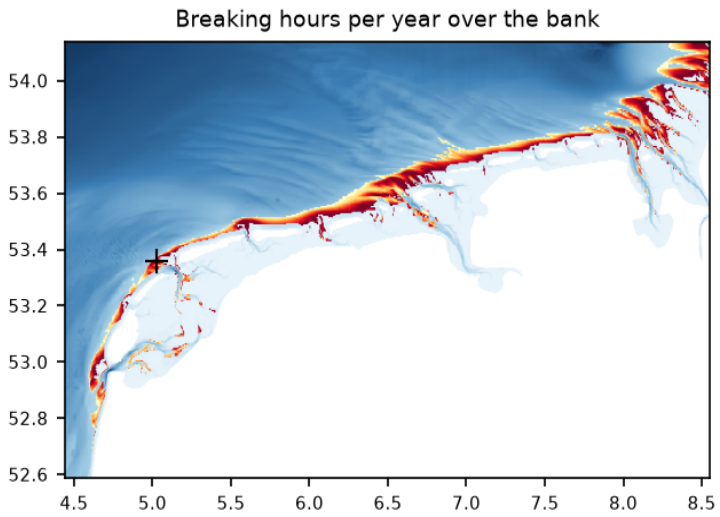


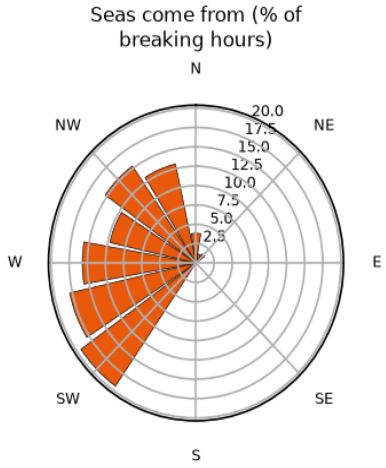


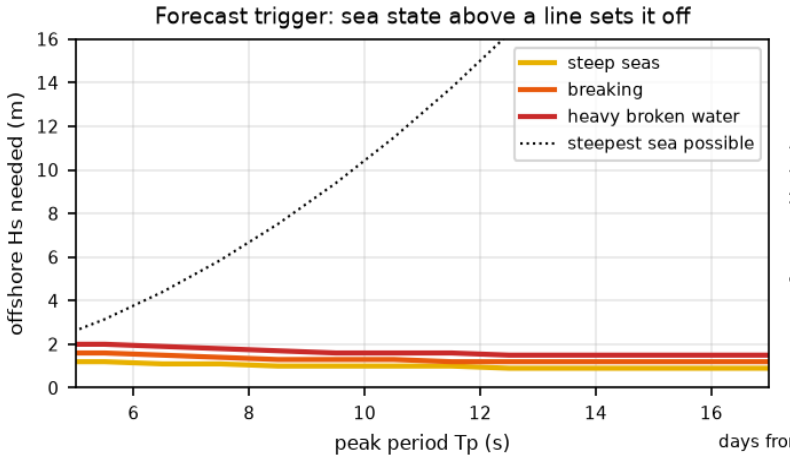


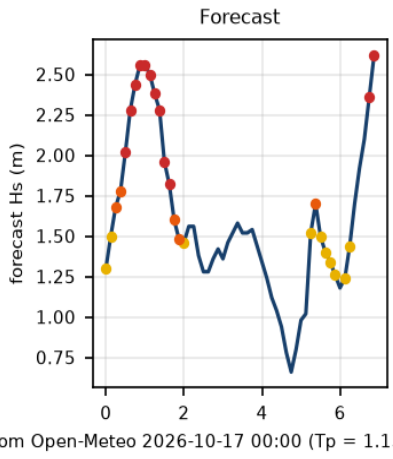
days from Open-Meteo 2026-10-17 00:00 (Tp = 1.13 x mean period)

62. Shoal 19 km off Terschelling (53.358 N, 5.027 E). Shallowest 3.0 m at low tide, 5 km offshore, 1489 km2 of breaking ground. About 2598 hours a year of breaking seas in 40.1 storms, mostly in Jan, Dec, Nov. Waves break when the offshore sea reaches Hs 1.4 m at Tp 8 s; Hs 1.3 m at Tp 10 s; Hs 1.2 m at Tp 12 s; Hs 1.2 m at Tp 14 s, from SW, WSW, W, NW. Worst in the record: 2022-02-18, Hs 5.4 m, Tp 10 s from WSW.

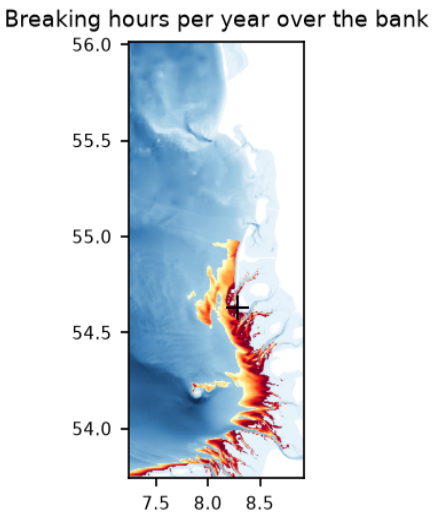


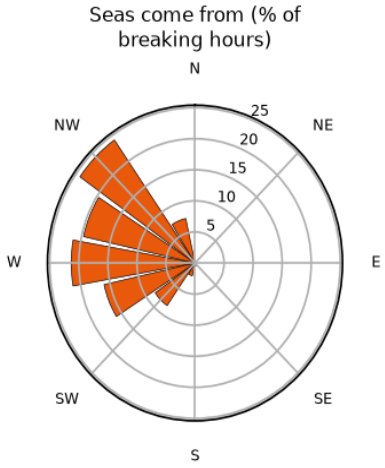


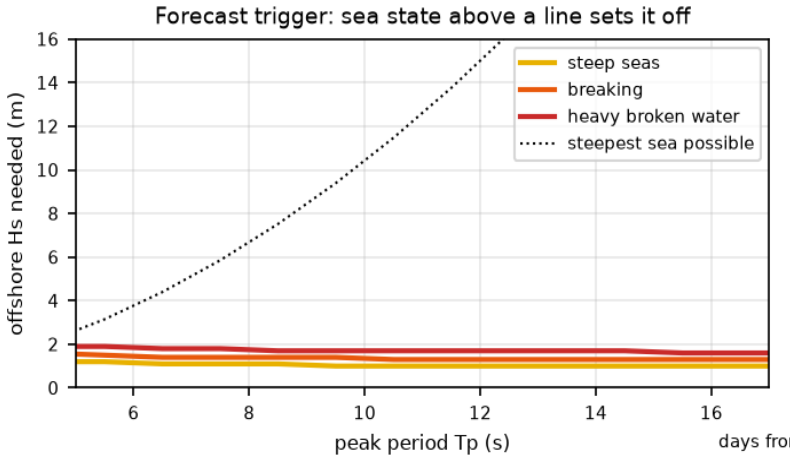


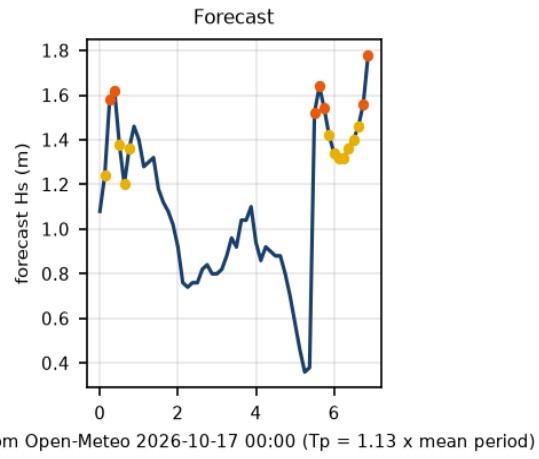


63. Shoal 30 km off Sylt (54.633 N, 8.287 E). Shallowest 3.0 m at low tide, 3 km offshore, 4101 km2 of breaking ground. About 2248 hours a year of breaking seas in 41.3 storms, mostly in Jan, Dec, Nov. Waves break when the offshore sea reaches Hs 1.4 m at Tp 8 s; Hs 1.4 m at Tp 10 s; Hs 1.3 m at Tp 12 s; Hs 1.3 m at Tp 14 s, from NW, W, WNW, WSW. Worst in the record: 2013-12-05, Hs 7.1 m, Tp 12 s from WNW.









How it works

Seabed. EMODnet's mean depth grid was cut into shallow cells (under 45 m and at least 2 km from the coast). For each cell the compass directions open to the sea were found: a direction is blocked if land or a drying bank lies within 25 km up-wave, which keeps sheltered water such as the Wadden Sea from scoring. Ground shallower than 3 m at low tide is left out: drying flats break in almost any sea and are charted no-go areas, so they are not a forecasting question. The Channel and the Norwegian skerry coast are left out too.

Breaking. The surf-finder broke a single swell at 0.78 times the depth. Ships meet a random sea, so this version uses the Battjes-Janssen model: waves in a sea state follow a Rayleigh spread of heights, and the depth and wavelength cap the largest wave the water can hold. The fraction above that cap breaks. Before reaching the bank the sea shoals (group-velocity shoaling from the depth the wave model sees to the bank). Three levels: steep (1 in 200 waves break), breaking (1 in 20) and heavy broken water (1 in 5).

Triggers. Inverting the model gives, for every cell and peak period, the offshore Hs that reaches each level. Periods where that Hs would be steeper than real seas get (Hs above a fifteenth of the deep-water wavelength) are marked as impossible.

Climate. 25 years of ERA5 waves every 3 hours were compared with each cell's triggers, counting only seas from its open directions. Cells that break at least once were grouped into banks. Ranking favours hours of breaking seas, heavy broken water, size, distance offshore and ship traffic (EMODnet vessel density). Wave history was only downloaded at 22 points this time, so these statistics cover banks within 110 km of them. Elsewhere a bank is listed if a severe storm sea (Hs 9 m, Tp 12 s) would break on it, and it is ranked by the Hs that sets it off.

Refraction. The surf-finder ray tracer was run over the top banks for the storm direction and period that most often sets them off, showing where the wave energy focuses and where it breaks first.

Limits. This is a screening study, not a navigational product. Tidal currents are not included, and wind against tide over a bank makes seas steeper and break earlier than shown. The 500 m grid misses small pinnacles and wrecks, sandbanks move, and ERA5 at 0.5 degrees underestimates the very biggest storm peaks. Bank names come from an approximate gazetteer and should be checked on an Admiralty chart.